

Lab 04

Due Feb 18 5:00 PM PST

AUTHOR

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Modify this template file to create your submission.

Problem 1a

```
my_lm <- function(x, y) {  
  
  x_mean <- mean(x)  
  y_mean <- mean(y)  
  
  a <- sum((x - x_mean) * (y - y_mean)) / sum((x - x_mean)^2)  
  
  b <- y_mean - a * x_mean  
  
  return(list(a = a, b = b))  
  
}
```

Problem 1b

```
## write answer here  
  
x_vec <- c(3, 0, 1)  
y_vec <- c(4, 3, 4)  
  
my_lm(x_vec, y_vec)
```

```
$a  
[1] 0.2857143
```

```
$b  
[1] 3.285714
```

```
model <- lm(y ~ x, data = data.frame(x = x_vec, y = y_vec))  
alpha <- summary(model)$coefficients[2, 1]  
beta <- summary(model)$coefficients[1, 1]  
  
alpha
```

```
[1] 0.2857143
```

```
beta
```

```
[1] 3.285714
```

Problem 2a

```
## write answer here
```

```
library(RSQLite)
library(DBI)
```

```
db <- dbConnect(SQLite(), dbname = "xcoretail.sqlite")
```

```
month_price <- dbGetQuery(db, "SELECT UnitPrice FROM prices WHERE StockCode == 71053")
price <- month_price[1,1]
price
```

```
[1] 3.39
```

```
months <- c("2011-01", "2011-02", "2011-03", "2011-04", "2011-05", "2011-06", "2011-07",
month_count <- numeric(8)

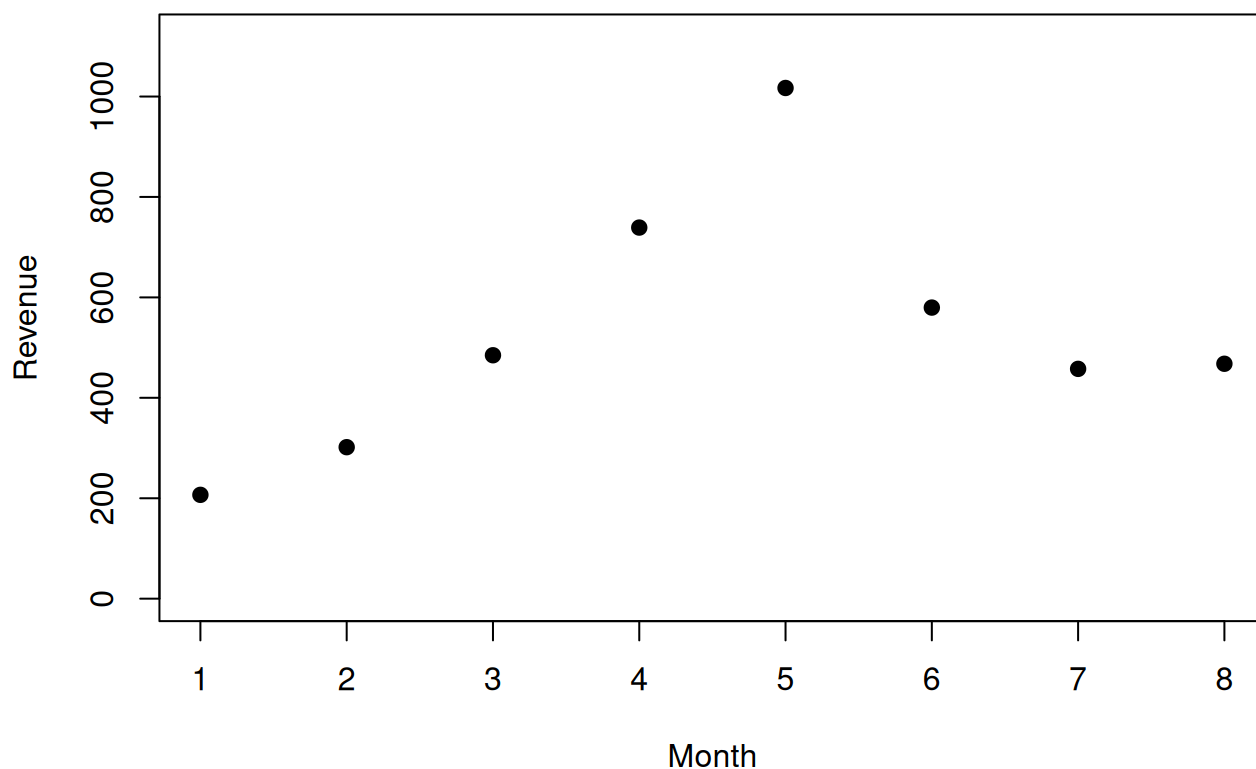
for (i in 1:8){
  query <- sqlInterpolate(db, "SELECT SUM(Quantity) FROM sales WHERE StockCode = 71053 AND
  month_count[i] <- dbGetQuery(db, query)[1, 1]
}
month_count
```

```
[1] 61 89 143 218 300 171 135 138
```

```
month_rev <- month_count * price
```

```
plot(1:8, month_rev,
     main = "71053 Rev",
     ylab = "Revenue",
     xlab = "Month",
     pch = 19,
     ylim = c(0, max(month_rev) * 1.1 ) )
```

71053 Rev



Problem 2b

write answer here

```
months <- c("2011-01", "2011-02", "2011-03", "2011-04", "2011-05", "2011-06", "2011-07",  
tot_month_rev <- numeric(8)  
  
for(i in seq_along(months)){  
  query <- sqlInterpolate(db, "SELECT SUM(s.Quantity * p.UnitPrice) AS Revenue FROM Sales  
  tot_month_rev[i] <- dbGetQuery(db, query)[1,1]  
}  
tot_month_rev
```

```
[1] 752584.2 606180.3 1358625.2 605242.6 863603.7 833007.0 845379.7  
[8] 832169.0
```

```
plot(1:8, tot_month_rev,  
     main = "Montly Rev",  
     ylab = "Revenue",
```

```
xlab = "Month",  
pch = 19,  
ylim = c(0, max(tot_month_rev) * 1.1 ) )
```

